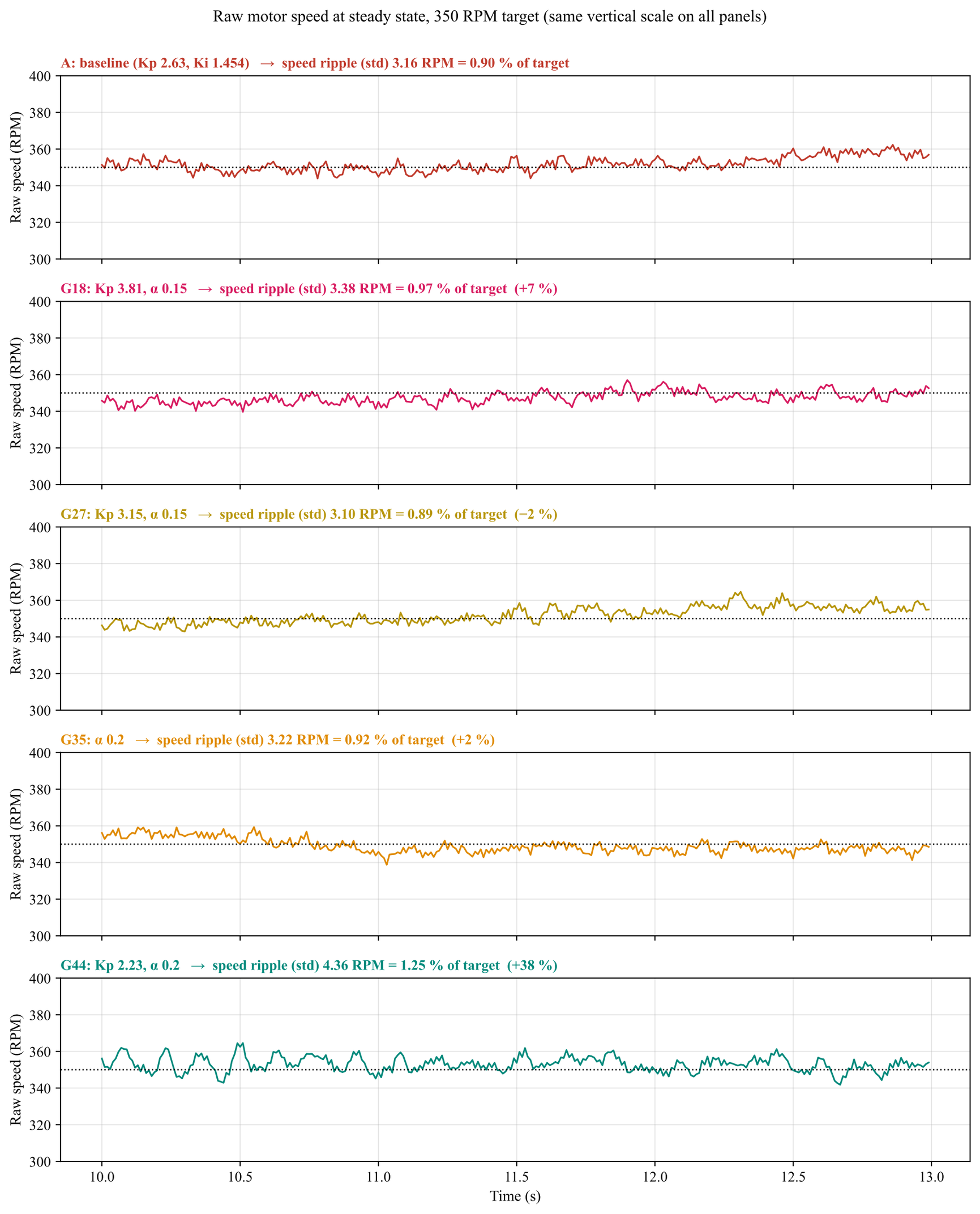
Raw motor speed at steady state, 350 RPM target (same vertical scale on all panels)
A: baseline (Kp 2.63, Ki 1.454) → speed ripple (std) 3.16 RPM = 0.90 % of target
G18: Kp 3.81, α 0.15 → speed ripple (std) 3.38 RPM = 0.97 % of target (+7 %)
G27: Kp 3.15, α 0.15 → speed ripple (std) 3.10 RPM = 0.89 % of target (−2 %)
G35: α 0.2 → speed ripple (std) 3.22 RPM = 0.92 % of target (+2 %)
G44: Kp 2.23, α 0.2 → speed ripple (std) 4.36 RPM = 1.25 % of target (+38 %)
Raw speed (RPM)
400
380
360
340
320
300
10.0
10.5
11.0
11.5
12.0
12.5
13.0
Time (s)